g CO2 stored /
MJ(LCV)
Emissions credit equivalent to the net CO 2 captured and
stored (long-term: 100 years)
gCO2eq /
MJ(LCV)
Emissions associated with the process of capturing,
compression and/or cooling and temporary storage of the CO2
gCO2eq /
MJ(LCV)
Emissions associated with transport to a long -term storage
site
gCO2eq /
MJ(LCV)
Any emissions associated with the process of storing (long -
term: 100 years) the captured CO 2 (including fugitive
emissions that may happen during long -term storage and/or
the injection of CO2 into the storage)
gCO2eq /
MJ(LCV)

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SOURCE: MEPC.376(80).pdf
MJ(LCV)
Any emissions associated with the process of storing (long -
term: 100 years) the captured CO 2 (including fugitive
emissions that may happen during long -term storage and/or
the injection of CO2 into the storage)
gCO2eq /
MJ(LCV)
Any additional emissions related to the CCS

4.5 The WtT emissions in Equation (1) include emissions associated with raw materials extraction or cultivation, primary energy sources used for production of goods and utilities such as energy carriers (e.g. fuels and electricity), transport and distribution (including bunkering), direct land use change and changes in carbon stocks (soil carbon accumulation).

4.6 Processing incorporates all steps and operations needed for the extraction, capture or cultivation of the primary energy source. Process includes basic transformation at source and operations needed to make the resource transportable to the marketplace (e. g. drying, chemical/physical upgrade such as gas-to-liquid, etc.).

4.7 Transportation, processing and distribution include transportation of the products in the fuel pathway to the place of transformation, conditioning (such as compression, cooling),

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SOURCE: MEPC.376(80).pdf
chemical/physical upgrade such as gas-to-liquid, etc.).

4.7 Transportation, processing and distribution include transportation of the products in the fuel pathway to the place of transformation, conditioning (such as compression, cooling),

distribution to the marketplace (i.e. bunkering) and eventual leakages, as well as fugitive emissions at any of these stages.

4.8 Allocation of emissions to co-products based on their energy content should be used, as the most appropriate and reliable methodology considering the establishment of an appropriate certification method using values that are predictable, reproducible and stable.

4.9 Land use (direct and indirect) for the production of biofuels may lead to land use change (LUC). LUC can be classified as direct LUC (DLUC) and indirect LUC (ILUC).

4.10 The DLUC definition is based on ISO 14067:2018 described as a change in the use or management of land within the product system being assessed. The DLUC impacts

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4.10 The DLUC definition is based on ISO 14067:2018 described as a change in the use or management of land within the product system being assessed. The DLUC impacts comprises the emissions and sequestration resulting from carbon stock changes in biomass, dead organic matter and soil organic matters, evaluated in accordance with the IPCC Guidelines. When available, sector or country -specific data on carbon stocks may be used; otherwise, IPCC's Tier 1 default emission factors may be considered. Two terms in the W tT Equation (1) capture respectively emissions resulting from direct land use change, i.e. el, and sequestration or otherwise increase in the content of soil organic carbon: esca.

4.11 The ILUC definition is based on ISO 14067:2018, described as a change in the use or management of land, which is a consequence of direct land use change, but which occurs outside the product system being assessed. ILUC occurs as a result of the economic impacts induced by increased biofuel demand on commodity prices with resulting shifts in demand and

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SOURCE: MEPC.376(80).pdf
outside the product system being assessed. ILUC occurs as a result of the economic impacts induced by increased biofuel demand on commodity prices with resulting shifts in demand and supply across economic sectors, including primarily food and feed production. ILUC cannot be directly measured and is projected with economic models instead.

4.12 Owing to the variability of assumptions underlying the evaluation of indirect effects, quantitative assessment of GHG effects of ILUC is subject to uncertainty, high quantitative variability and to the risk of arbitrary conclusions. For these reasons, ILUC shou ld be at this stage addressed using a risk -based approach, meaning that quantitative values will not be calculated and assigned to each fuel pathway. The ILUC emissions, as well as the spatial dimension of the ILUC effects, are dependent on a var iety of factors such as local/regional

conditions and practices for agriculture, current and expected food import demand, national current accounts, the type of feedstock, the alternative economic uses of the same feedstock, etc.

4.13 A qualitative risk-based approach to ILUC includes consideration on the following:

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SOURCE: MEPC.376(80).pdf
current accounts, the type of feedstock, the alternative economic uses of the same feedstock, etc.

4.13 A qualitative risk-based approach to ILUC includes consideration on the following:

.1 Low-ILUC risk qualifies and characterizes biofuel production projects that supply additional feedstock without disrupting existing land uses. When productivity is increased on an area which is in agricultural production, only additional yields should be considered as low-ILUC rather than the entire production; and

.2 High-ILUC risk qualifies and characterizes biofuel production projects based on, or displacing, food and feed crops resulting in a significant expansion of the feedstock production area shifting into land with high-carbon stock.

4.14 WtT default emission factors are provided in appendix 2 of these guidelines.

5 TANK-TO-WAKE (TtW)

5.1 The aim of the TtW methodology is to quantify and evaluate the intensity of CO2, CH4 and N2O emitted on board a ship related to the fuel usage, including combustion/conversion and all relevant fugitive emissions with a Global Warming Potential.

RESOLUTION MEPC.376(80) (adopted on 7 July 2023)

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and N2O emitted on board a ship related to the fuel usage, including combustion/conversion and all relevant fugitive emissions with a Global Warming Potential.

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5.2 The TtW GHG emission factors should be calculated using Equation (2):

Equation (2)

= 1 ((1 1100 (_ + C)) × (2 × 2 + 4 × 4 + 2 × 2) + + (1100 (_ + C) × ×) × ×)

Note: terms , and are pending further methodological guidance to be developed by the Organization. For more details refer to footnotes 11 to 13.

Term Units Explanation
Cslip_ship

% of total fuel
mass
Factor accounting for fuel (expressed in % of total fuel mass delivered to the ship) which escapes from the energy converter without being oxidized (including fuel that escapes from combustion chamber/oxidation process and from crankcase, as appropriate)
_ = (1 /100)
Cslip

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SOURCE: MEPC.376(80).pdf
delivered to the ship) which escapes from the energy converter without being oxidized (including fuel that escapes from combustion chamber/oxidation process and from crankcase, as appropriate)
_ = (1 /100)
Cslip